Integrating Unordered Time Frames in Neural Networks: Application to the Detection of Natural Oil Slicks in Satellite Images

Antoine Scardigli, Laurent Risser, Chihab Haddouche, and Romain Jatiault

***Abstract*— In this article, we explore the use of novel neural-network architectures to distinguish natural seepages from artificial slicks in synthetic aperture radar (SAR) images. They exploit a distinctive property of natural seepages, which is their temporal recurrence in the same geographical area. This information can be captured in different SAR images acquired at the same location over time, but not necessarily at a regular time frequency. The proposed neural-network architectures are then built as specific block layers, which efficiently treat the unordered temporal information, followed by more conventional neural-network layers, which are widely used for image classification. Different block layers for unordered temporal information are compared on Sentinel-1 images acquired in the Aegean Sea. Following data augmentation steps, our dataset contains a consistent subset of images gathered among 16 000 time-patches. We demonstrate that nonlinear time-based block layers and block layers that avoid information bottlenecks are most efficient to discriminate between natural and artificial oil spills. Compared with standard neural-networks which use single time frames, the integration of unordered temporal information increases the overall accuracy (OA) from 82% to 92% on our dataset, which demonstrates the effectiveness of the proposed approach.**

***Index Terms*— Neural networks, oil seeps detection, remote sensing, synthetic aperture radar (SAR), temporal information integration.**

I. Introduction

THE presence of oil at the sea surface can be associated with two main sources. First, liquid hydrocarbons may be pollutions originating from oil platforms or ships occurring during the extraction or transportation processes. Second, oil naturally escaping from the sedimentary series can generate a consistent volume of oil in the water column and at the sea surface [1], [2]. Most sedimentary basins were documented to show evidence of natural seepages [3], [4]. It is estimated that the volume associated with natural seepage is equivalent in volume to that of pollutions. Inventorying and quantifying the amount of natural oil seeps detectable at the sea surface appears as important to understand their environmental effect and, ultimately, to integrate this information into the global carbon budget.

The presence of hydrocarbons at the sea surface reduces the amount of backscattered energy on radar images from synthetic aperture radar (SAR), which makes hydrocarbons easily detectable under certain wind and angle conditions [5]. The discrimination between natural seepages and other types of surfactants (look-alikes, i.e., surface microlayer, [6], [7], [8]) is essential to provide reliable estimations of naturally expelled volumes [1]. Natural seepage slicks follow sea surface currents and winds, which creates curvilinear shapes with gentle angles [2], [9], [10] or circular shapes [11]. The textures, sizes, shapes, or context (presence of boats or platforms in proximity) can be used as a first interpretation hint by the human operator to distinguish natural seepage compared to other sources of slicks.

Multiple integration methods have been considered to refine oil slicks sources discrimination, including polarimetric studies, sea surface ocean modeling, and winds assessment [10], [12], [13], [14], [15], [16]. Those methods rely on the integration of miscellaneous data types for which the reproducibility is eventually restricted to concomitant data availability. The motivation of this work is to propose an alternative approach for oil slicks sources discrimination that aim to be widely reproducible and only relies on remotely acquired data.

Natural oil seeps are emitted from a fixed source at the seafloor [17], [18], the migration oil droplet may be affected by a lateral deflection in the water column [19], [20] but these deflections remain low compared to the water depth [21]. The oil-surfacing area [22] always occurs in a local area (typically circles with diameters lower than the water column) between successive slicks, which create typical diverging patterns on the slicks compilation map composed by individual oil slicks expelled from a fixed emission point. Additional anthropogenic sources, such as sunken ships or oil platform emitting oil production waters may create similar recurrent oil slicks patterns. Oil-emitting platforms (RIG or FPSO) [23]

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Antoine Scardigli is with Ecole Normale Supérieur (ENS) Paris Sciences et Lettres University, 75005 Paris, France.

Laurent Risser is with CNRS, Institut de Mathématiques de Toulouse, 31400 Toulouse, France.

Chihab Haddouche and Romain Jatiault are with the Centre de formation et de recherche sur les environnements méditerranéens (CEFREM, UMR5110), CNRS, University of Perpignan Via Domitia, 66100 Perpignan, France (e-mail: romain.jatiault@univ-perp.fr).

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can be dismissed with natural oil seepages considering the diffracting point associated with the platform location. Similar patterns may however be generated by frequently oil-leaking shipwreck even if most are referenced among worldwide database [24]. Apart from these two specific cases, oil slicks reoccurring in the same area is a distinctive property of oil seeps compared with oil spills or look-alikes and can be used as a discriminative technique. The multitemporal recurrence can be revealed during the interpretation of the multitemporal dataset.

Heliosynchronic satellites periodically cover the same area, and the revisiting time over the Mediterranean sea is particularly high, which enables a high data coverage. The volume of daily Sentinel-1 acquisitions (8 TB) is too high to provide manual interpretations of oil slicks delineations, so this process needs automation systems.

Automatic oil spills detection and classification from SAR images was historically performed through an empirical and expert-made feature extraction of SAR images, where the features would be given to a classifier. Solberg and Solberg [25] used for the first time a machine learning technique (decision trees) for the oil spill detection task. This classifier however relied on insightful expert-made features, such *distance to nearest bright object*, assuming that the nearest bright object would be the polluting boat. Kubat et al. [26] used various machine-learning classifiers, and more recently, Garcia-Pineda et al. [27], [28] used fully connected neural networks with one hidden layer as a classifier, but still used a separated manual feature extraction process [29], [30].

CNNs were first introduced in 1998 by LeCun [31], [32], and [33] first used convolutional neural networks (CNNs) on SAR oil images, coupled with some empirically extracted features, to discriminate oil slicks from look-alikes. For the first time, the classifier participated in the feature extraction at the cost of interpretability.

The wavelet-neural-network method (WNN) [34], [35] led to a 5% to 7% accuracy improvement in oil spill detection. Liu et al. [36] showed that the CNN from [32] outperformed simpler ML approaches such as SVM or Random Forest. Using VGG16 [37] architecture on a giant dataset for oil spill detection, Zeng and Wang [38] reached an accuracy of 94%. The authors explained that the deep neural network learned to extract features of significantly higher quality than those empirically built by experts.

Deep CNNs (DCNNs) were used in the oil spill segmentation field before being used in the oil classification and oil detection field: Fully connected networks (FCN) [39] have been shown to outperform standard machine learning algorithms such as logistic regression or K-nearest neighbors (KNN) [40]. Bianchi et al. [41] compared several segmentation models and concluded that the best algorithm for her segmentation task was FCN. Krestenitis et al. [42] used DCNN for oil segmentation from SAR images and compared different state-of-the-art algorithms in this context: DeepLabv3+ [43], DeepLab [44], and UNET [45], [46], and revealed that DeepLabv3+ is the best segmenting approach for his task. The current state-of-the-art in oil spill segmentation is a 91% intersection over accuracy score reached by Zhang et al. [47]. The superiority of deep neural networks for oil slicks extraction however remains unelucidated because very recent approaches still use manual feature extraction combined with a historical classifier in an SAR oil classification studies [48].

To distinguish natural from anthropic oil slicks, feature selection with machine learning tools was performed on more than 400 features manually extracted and reached an 82% accuracy in natural/anthropic oil distinction using a random forest classifier [49]. This is the state-of-the-art with which we will compare our approach.

As far as the authors know, all existing machine learning approaches for oil slick detection in remote sensing are based on single mono-temporal images. For the first time, our article focuses on spill/seeps distinction using a multitemporal approach to discriminate redundant natural seepage features in the automatic detection process. This methodological contribution is motivated by two facts: first, natural seepages are temporally recurrent in the same geographical area. In addition, this information can be captured in different SAR images acquired at the same location over time. It would then be tempting to use convolutional layers on the temporal dimension of the different images acquired at a given location to extract temporal features that would help detecting this recurrent phenomenon. This technique is indeed at the heart of DCNNs, which first extract local image features out of the input images and then use these image features for classification, regression, or dimensionality reduction tasks. This approach is however not relevant in our application, as natural seepages are temporally recurrent but not necessarily at a regular time frequency. It can also be remarked that the image acquisitions are also generally irregular in time. Our methodological contribution is then specifically to propose and to explore the use of specific block layers in neural networks, which efficiently treat the unordered temporal information. These block layers therefore ingest the multitemporal characteristics of natural oil seeps, and then use the resulting information as a distinctive criterion to discriminate anthropogenic oil spills.

II. Material and Methods

A. Data

1) Sentinel-1 Images: The dataset used in this study is composed of Sentinel-1 products provided by the European Space Agency (ESA). The Sentinel-1 satellite onboard a SAR system, which is a side-looking active sensor. This system has long been demonstrated to be highly efficient for the detection of oil slicks at the sea surface [6], [8] even for the thinnest slicks [1]. The thickness detection threshold is 0.1 μm [22]. The SAR scenes acquisition mode consists of a dual-polarimetric system (IW mode of Sentinel-1) combining the vertical transmit, the vertical receive (VV), the vertical transmit and the horizontal receive (VH) polarization channels. We selected the VV polarization mode due to the reported higher detection performance and higher signal-to-noise ratio [50]. The spatial resolution of the IW mode

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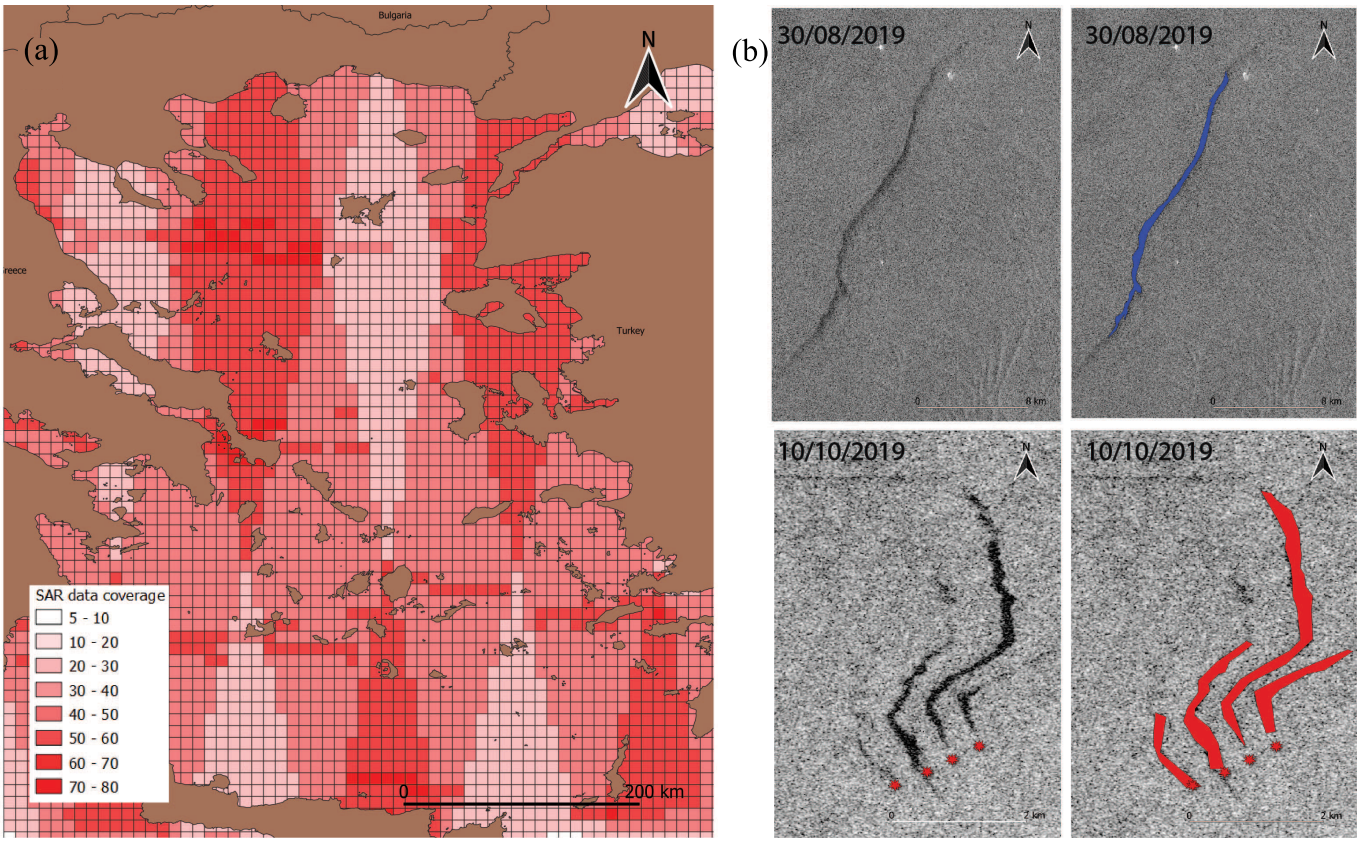

Fig. 1. Sentinel-1 data used for the detection of natural oil slicks in satellite images. **(a)** SAR coverage of the images used in our tests. They were acquired in the North Aegean Sea during the summer periods of 2018 and 2019. **(b)** Extract of Sentinel-1 satellite images showing the two labels used in this study (oil spill related to mankind activities in blue; natural oil seepage in red) with and without manual labeling.

of Sentinel-1 is 20 m resampled at 10 m. The dataset is composed of 488 GRD processed scenes acquired during the summer periods of 2018 and 2019 (from June to September) to ensure similar meteorological conditions throughout the dataset but also to avoid unsuitable conditions associated with strong winds during winter periods. Both ascending and descending modes were collected and the incidence angles typically ranged from 29° to 46°. SAR scenes were calibrated, multilook processed, ellipsoid and terrain corrected.

As shown Fig. 1, the number of visits during the observation period is highly variable from place to place across the study area, depending on the acquisition configurations. These 488 satellite scenes overlap between 30 and 60 times depending on the exact location across the North Aegean Sea. The data coverage varies from 20 to 80 stacked images. The selected study area corresponds to the Northern part of the Aegean Sea, which corresponds to a confined marine environment with water depths reaching 1500 m for marine through.

2) Manual Labeling: The detection of oil slicks is based on a visual recognition of satellite images followed by a digitisation step, which consists of outlining of the slicks using a geographic information systems (GIS) software. Each polygon is then labeled depending on the interpretation of the origin and a ranking is provided regarding the confidence of the interpretation origin. Natural oil seepage or pollution slicks are visually discriminated based on typical geometries detected by the photo interpreter. The shape of natural oil slicks is controlled by the velocity and the direction of the wind and surface currents [8], which are highly variable compared to deeper water bodies displacements. This induces curvilinear shapes for natural seepage slicks [2].

Compared with other phenomena also reducing the volume of energy backscattered to the satellites (similar radar signatures to oil slicks), natural seepage is a repetitive phenomenon that occurs in similar areas over subsequent satellite acquisitions. This criterion is especially discriminative and easily detectable on time series. The contouring of oil slicks is then manually performed using geographical information system (GIS) software and stacked along the interpretation process of satellite images.

The dataset size of such studies is usually limited to a few hundred of images due to the necessary effort to ingest and interpret the data. The training dataset we used in our experimental setting therefore is inline with typical oil slicks studies datasets and is representative of the operational conditions. In the context where a massive amount of annotated training data is not available and where the classification can be biased by subtle artifacts, we believe that using deep neural-network architectures is not necessarily a pertinent choice because it requires a large amount of labeled data. This prompted us to explore the use of specific neural network architecture. We are aware that this specificity somehow implies strong assumptions on the properties, but we believe that it has great potential to improve the accuracy of predictions.

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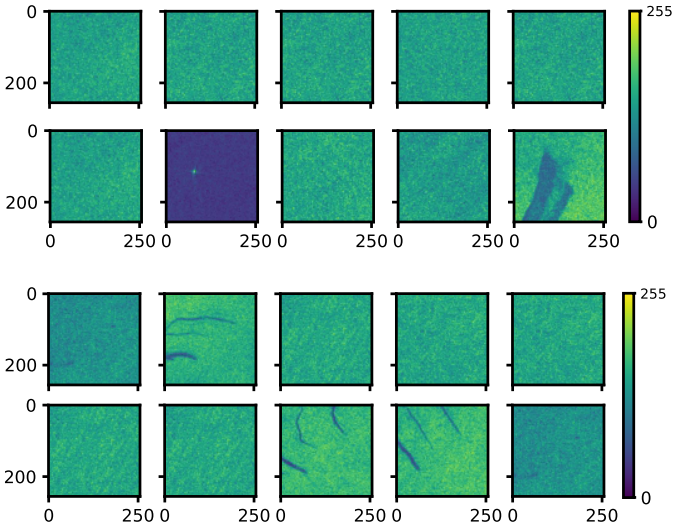

Fig. 2. (Top) Time-patch containing an artificial oil spill. Note that a single oil film is observed among all patches. (Bottom) Time-patch containing natural oil seepages. Importantly, the images composing the time-patches were acquired at pseudorandom times, which suggests that taking into account the unordered recurrence of oil patterns observed in the images may be pertinent to distinguish artificial oil spills from natural oil seepages.

3) Preprocessing: Raster images and their corresponding segmentation were divided into subsets (called patches) with a fixed size of 256 × 256 pixels in agreement with DCNNs input requirements: small inputs with constant size.

Overlapping patches that covered the same area but at different times were stacked to generate time-patches (see the examples Fig. 2).

To obtain time-patches, we used the latitude and longitude top left GPS coordinates of the raster images to cluster them into groups of images with more than 95% of shared area covering. Since the satellite is helio-synchronic, it periodically covers the same areas, and such overlaps are relatively frequent. The clustering resulted in 56 groups of images with similar pixel ground print. As the data coverage is not homogeneous (Fig. 1), the dimensions of time clusters were variable. In our case, the average size of each time-patch was eight images. Only 32% (18) of all clusters had more than ten images.

Images within the same cluster were then cut and cropped such that they were aligned at a pixel level. We applied a threshold noted c regarding the number of patches along the time dimension. Clusters with less than c images were discarded, and we kept the c images with the largest number of oil spill pixels from clusters with more than c images. The c time-aligned images were finally gathered into time-patches of dimension $c \times 256 \times 256$, through the application of a grid that crosses the whole time-image. In our study, c equals 10. This value was chosen using grid search for all values of c between 3 and 30, and was found to be the best compromise between the quantity of data available and how easy is the discrimination task.

As our goal is to classify the time-patches containing natural seepages from those containing pollution spills, we only used time-patches containing at least one of these events (based on the manual segmentations of the corresponding images). We also removed the few time-patches where both events were simultaneously segmented as they were infrequent.

For the baseline approach that uses normal patches without time superposition to perform the classification task, we only kept the patches containing manually segmented natural or artificial oil spills.

B. Methods

1) Overview of the Neural-Network Architectures: Our methodology relies on the use of specific block layers, which efficiently treat unordered temporal information, in Neural Networks. The general proposed Neural Network architecture, that we denote *time-approach*, is illustrated Fig. 3: The block layers treating the unordered temporal information first integrate a set of c time-patches acquired at the same geographical location, but at different times. The transformed time-patches are then incrementally treated by more conventional neural network layers, which we denote baseline neural network architecture.

It is important to note that the novel use of these block layers to address unordered temporal information is motivated by the fact that the acquisition time steps may be subject to strong fluctuations and that the temporal order of the slick detection is not a criterion determining their origin.

2) Selection of a Baseline Neural-Network Architecture: The baseline and the time-approach use the same baseline neural network architecture and the same dataset. The baseline approach however directly uses the individual patches of the dataset with their corresponding segmentation, and the time-approach uses time-patches with unified segmentations. By referring to Fig. 3, the baseline approach would then differ from the time-approach by the fact it inputs a single patch and that it skips the *Integration of temporally unordered patches* step.

It is interesting to note that DCNNs often aim to classify colored images. The first layer of such neural networks therefore takes patches with three channels as input, where the three channels are classically red, green, and blue. In our case, a channel represents a greyscale image of 256 × 256 pixels in a given area. All c channels represent this area at different times, with no hypothesis about their acquisition times. Input time-patches will then have dimensions $c \times 256 \times 256$ for the time-approach, and $1 \times 256 \times 256$ for the baseline approach.

To make the dimension of the input images of the baseline approach (one channel) match the dimension of the neural network inputs (three channels), a basic strategy is to replicate the single channel three times. A mathematically equivalent strategy is to alternatively redefine the first layer of a predefined neural network architecture with one input channel, functionality which is offered by PyTorch or Keras for instance.

To make the dimension of the time-patches (c channels) fit with DCNNs, the c channels can be compressed into three channels to gather our time-patches to the standard DCNNs. However, this first strategy might induce a consistent loss of information. Another strategy is to modify or shortcut the first layer such that DCNNs would accept inputs with

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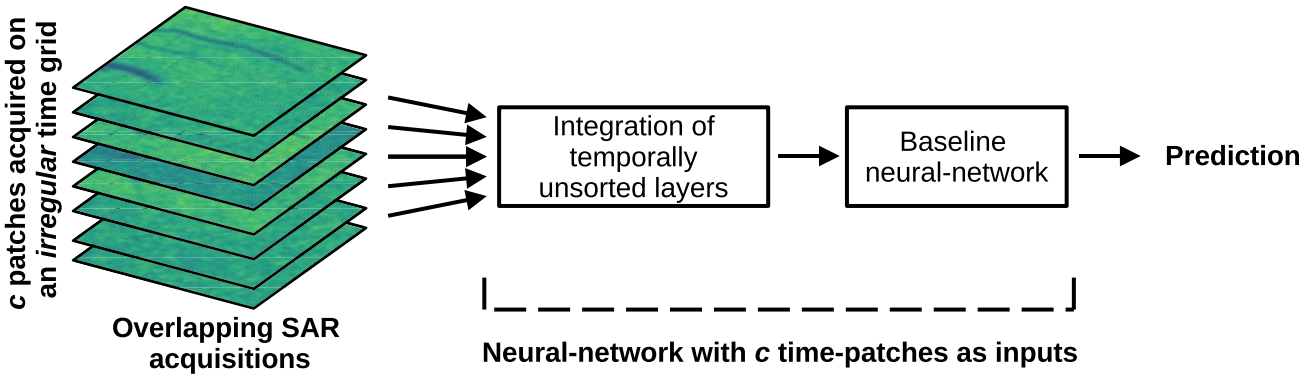

Fig. 3. Illustration of the *time-approach*. The input time frames representation is first transformed by the block layers integrating the unordered time information. A standard neural-network architecture is then used for the rest of the prediction.

c channels. Exploring and comparing different approaches among both strategies to address the temporal dimension is the main methodological contribution of this article. Indeed some recent work [51] use temporal dimension for monitoring of oil seepages, but we are the first to use it for oil spill/oil seepage distinction. We will compare in this sub-section four DCNNs on the baseline task, and then extend in the next subsection the best of these DCNN with our time-approach task. The reason why we only use DCNN and no Recurrent Neural Network, which are known for treating time series is that we are not handling time series (that have time relations) but sets of images. The literature recommends using CNNs because CNNs consider the inputs as unordered.

The four compared DCNNs are VGG11, VGG19_bn, RESNET18, and RESNET152. VGG11 and VGG19 belong to the family VGG [37] of DCNN. VGGs made a breakthrough in terms of top5 accuracy on the ImageNet dataset [52] in 2014 by reaching 92.5% top5-accuracy when the state-of-the-art was 86.3%. It is characterized by a long succession of 3×3 convolutions without filters and uses a huge number of parameters: several hundreds of millions parameters. This number of parameters is still modest compared with equivalent fully connected networks. VGG11 has 11 such layers and VGG19_bn has 19 layers and uses batch normalization between every layer to reduce the risk of overfitting.

ResNet networks were the first to perform better than humans on the ImageNet dataset by reaching 94.3% top-5 accuracy [32]. Its architecture is similar to VGG except that some connections may skip a layer, making it possible to mix or to select filtered and nonfiltered information. Architectures inspired by ResNet are often used in image classification [53]. ResNet18 has 18 layers and Resnet 152 has 152 layers. In practice, the DCNN requires a minimum number of layers to capture all the complexity of the task and a maximum number of layers to avoid overfitting [54], i.e., to avoid specializing too much on the training data without being able to generalize on test data.

To compare the efficiency of these baseline DCNNs approaches with other classic machine learning models, we will also measure the accuracy of the random-forest, SVM and K-nearest neighbors' models. The main difference between these models and DCNNs is that they consider all pixels as independent variables when making their prediction, while DCNNs take into account the proximity between pixels by design.

3) Extraction of Statistical Features: We will present in Section II-B4 the proposed block layers to integrate the unordered time dimension. To fairly compare these architectures in Section III with baseline neural network architectures taking a single channel as input, this single channel will possibly represent statistical features classically extracted out of the c channels. We then describe hereafter these statistical features.

We first define as Z a time-patch of dimension $c \times 256 \times 256$. The intensity $Z[k, i, j]$ is then obtained at the pixel $[i, j]$ of the kth patch of Z, and ranges between 0 (dark) and 255 (bright). We then introduce $f[i, j] = \mathrm{Var}(\{Z[k, i, j]\}_{k<c})$, where $\mathrm{Var}(\cdot)$ is the variance. The intuition behind this extractor is that natural seepages and artificial slicks can be distinguished in case several natural seepages occur at the same pixels timewise because this occurrence makes the variance higher. It should also discriminate between land and sea as they cause a very high and a small variance, respectively. It should finally be sensitive to lookalikes or boats.

We also introduce $g[i, j] = \mathrm{PAv}(\{N(Z, k)[k, i, j]\}_{k<c})$, where $\mathrm{PAv}(\cdot)$ is the power average function and $N(\cdot)$ is a normalization function. Specifically, we first use

$$\mathrm{PAv}(x_1, x_2, \ldots, x_n) = \left(\sum_{i=1}^{n} \left(1 - \frac{x_i}{\max_j (x_j)} \right)^{10} \right)^{10^{-1}}$$

that averages a list of pixels such that darker intensities have more weight than brighter intensities. Those dark intensities correspond to the pixels of oil slicks and oil seeps. The normalize function N also takes a channel as input and transforms its pixels intensities so that they follow a normal distribution. It is defined as

$$N(Z, k)[k, i, j] = \frac{Z[k] - \mu}{\sqrt{\frac{\sum_{i=0}^{255} \sum_{j=0}^{255} (Z[k, i, j] - \mu)^2}{256^2}}}$$

where $\mu = (\sum_{i=0}^{255} \sum_{j=0}^{255} (Z[k, i, j])^2)/256^2$. The function PAv averages the shapes of very dark objects while the pixel with light or average colors will be discarded as they will weight very little in the power average. Because PAv preserves

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the shape of the objects, lookalikes, natural, and artificial slicks can be further classified easily by the DCNN. The normalization channelwise into a normal distribution only assigns extreme values to pixels with values uncommonly distant from the mean. Hence, if there are fewer dark pixels in an image, these pixels will be assigned a relatively lower value. Because natural slicks are thinner than artificial slicks, and lookalikes are often wider than artificial slicks, the discrimination can not only use the shape of the objects, but also their values.

We finally define h such that $h[i, j] = \text{mean}(\{Z[k, i, j]\}_{k<c})$. Averaging the images should cancel noisy pixel values of the sea and conserve the shapes of oil seepages or slicks. In case several pixels contain oil seepages, the average will be lower than in the case of a single dark pixel from artificial spills.

4) Integration of the Unordered Temporal Information: We now present different block layers to integrate the unordered temporal information. We recall that they input time-patches of c channel and transform the representation of this information such that this transformed data representation is more discriminant than the original one for a classifier. We will see that these outputs will be 3 or c transformed time-patches. Those different block layers are denoted *time-methods* and their outputs are denoted *meta-inputs*.

a) Full convolution: The first time-method we present is to use a full convolution from dimension $c \times 256 \times 256$ to dimension $3 \times 256 \times 256$. This method applies a different convolution filter to all c channels, as described in [31]. This approach is illustrated in Fig. 4. An example of time-patch processed by this approach is presented Fig. 5. Detailed schemes of other time-methods presented in this article can be found in the Appendix.

b) Constrained convolution: The second time-method is to use a constrained convolution from dimension $c \times 256 \times 256$ to $3 \times 256 \times 256$. Constrained convolution applies the same convolution filter to all c input channels contrary to full convolution which applies a different convolution filter for all c input channels.

c) Averaging: This time-method consists of first splitting the c channels into three groups of the same size, averaging them, and selecting those three averaged groups as the three input channels. Note that this solution should not be scalable for a big c because the expected value of all pixels should be the same.

d) Statistical extraction: This time-method outputs three empirical statistical properties of the time-patches that are considered as meaningful. We will use in our results three statistical extractors out of Section II-B3. We note a, b, and c these statistical extractors. The metainput has then a shape $3 \times 256 \times 256$ and is the concatenation of three channels of dimension 256×256: $[a(Z), b(Z), c(Z)]$.

e) Sorting: In the *sorting* time-method, we output the three patches corresponding bitwise to the three smallest pixels timewise from the c patches. More formally it returns $\text{min3}_k Z[k, i, j]$, where min3_k is the function that outputs the three smallest elements of the k axis at each pixel $[i, j]$.

Since oil seepages, oil spills, and perturbations have typically low pixel intensities, the operator min3_k will keep only those and discard pixel values without interest out of the time-patches. This masking makes the discrimination much easier for the DCNN: for example, if only a single dark band appears over all the time-patches, the DCNN will very easily classify this shape as an oil spill. In the other case of an oil seepage, the DCNN will likely recognize characteristic dark interleaving thin band. Assuming that large perturbations occurring in a single patch have almost uniformly pixel values darker than oil, or lighter than oil, the recognition of interleaving thin dark bands can happen on the two remaining output channels.

This is illustrated in Fig. 6, where an example of a time-patch being preprocessed with the sorting time-method is shown. As defined in this paragraph, the pixels of the three augmented images [Fig. 6 (bottom)] contain the darkest intensities of the corresponding pixels through the time-patch. Specifically, the pixel intensities at the bottom-left, bottom-center, and bottom-right of Fig. 6 are respectively the darkest, second darkest, and third darkest intensities of the input time-patches.

f) Shortcutting full convolution: This time-method consists in shortcutting the first layer of the network and applying a full convolution directly from $c \times 256 \times 256$ to $64 \times 256 \times 256$.

g) Shortcutting constrained convolution: This time-method slightly differs from the previous one as it consists in shortcutting the first layer of the network and applying a constrained convolution directly from $c \times 256 \times 256$ to $64 \times 256 \times 256$.

h) Sorting + shortcutting full convolution: Here, time-patches are ordered along the time dimension, to give the c pixelwise-ordered images to the second layer of the NN, through a full convolution.

i) Sorting + shortcutting constrained convolution: In this last time-method, the time-patches are also ordered along the time dimension to give the c pixelwise-ordered images to the second layer of the NN, through a constrained convolution.

5) Expected Benefits and Constraints of the Proposed Time-Methods: We present in Table I a comparison of the benefits and constraints of each time-method, where the indices mean: [A]: Many learnable parameters; [B]: Different filters can be applied on every channel depending on their index; [C]: Information of c channels are reduced to three channels; [D]: Same filter is applied to all channels; [E]: Is linear both in the time dimension and the image dimension; [F]: Very simple; [G]: Does not scale well; [H]: Uses three diverse statistical extractors and are nonlinear in the time dimension and the image dimension; [I]: Nonlinear in the time dimension; [J]: Must choose only three statistical extractors; *[K]*: Does not exploit the image dimension; [L]: No information bottleneck; [M]: Linear in the image dimension.

6) Data Augmentation Considering the Temporal Information: For consideration of the c time-patches as a set by the neural network without rank relation (there is no reason that patches in the time-patch appear in a particular order), we introduce a novel data augmentation method which consists of applying a random permutation along the time axis of the c time patches. The possibility for the neural network to see $c!$ times more different time-patches will reduce the

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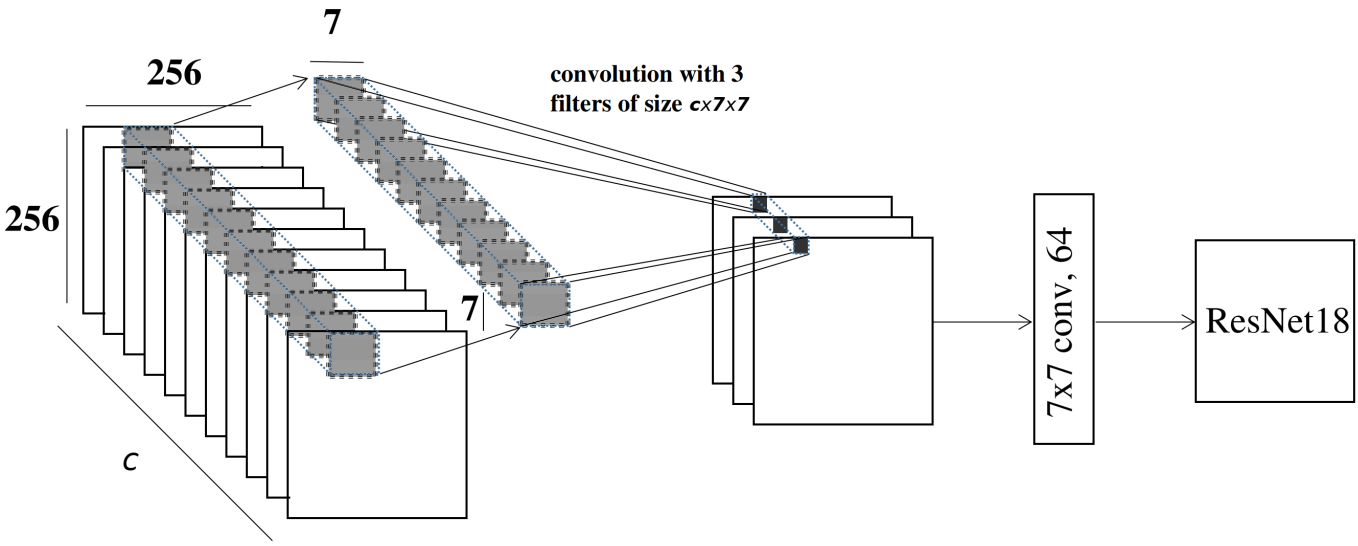

Fig. 4. Detailed temporal integration model for the full convolution time-method.

TABLE I

BENEFITS AND CONSTRAINTS OF THE APPROACHES FOR THE TIME DIMENSION INTEGRATION

Time method	Benefits	Constraints
Full conv.	[A]	[B] [C]
Constrained conv.	[D]	[C] [E]
Average	[F]	[C][E] [G]
Statistical extraction	[H] [I]	[C] [J]
Sorting	[I]	[C] [K]
Shortcutting full conv.	[A] [F]	[B] [E]
Shortcutting constrained conv.	[D] [F]	[E]
Sorting + Shortcutting full conv.	[A] [B] [I] [L]	[M]
Sorting + Shortcutting constrained conv.	[I] [L]	[D] [M]

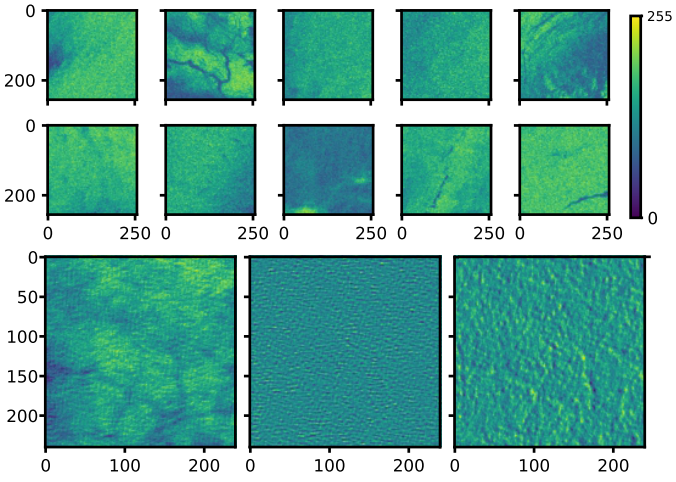

Fig. 5. (Top) Images out of a time-patch, and (Bottom) its corresponding meta-input after being processed by the full convolution time-method.

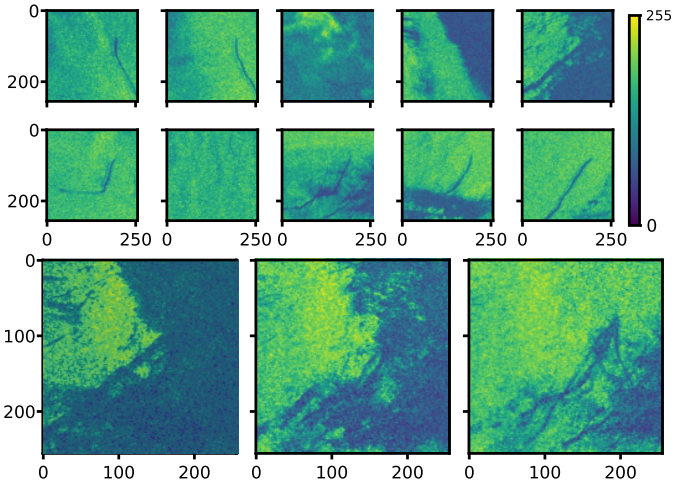

Fig. 6. (Top) Images out of a time-patch, and (Bottom) its corresponding meta-input after being processed by the *sorting time-method*.

overfitting effect (poor generalization on unseen data). This novel augmentation is performed before the application of the time-methods. Furthermore, we augmented the images differently each time they were accessed by the DCNN to avoid expensive data storage. We integrated traditional techniques such as cropping, rotations, flipping, adding noise, or zooming (See Fig. 6 for an example of data augmentation). Such augmentations were performed after the application of the time-methods. Assuming that the augmentation process is rich enough, this augmentation protocol expands the training dataset by a factor of the number of epochs.

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To assess that our method generalizes to unseen data, we use a train/validation/test dataset principle: we train on the training dataset, select hyperparameters on the validation dataset, and evaluate the methods on the test dataset. We split our dataset into the training dataset containing 80% of the images and on which the DCNN trains, the validation dataset containing 10% of the images and on which the DCNN can be evaluated to compare models, approaches, or hyperparameters, and the testing dataset that is only used on the remaining images for final evaluation of performances on unused data.

In our results, the training dataset contains an amount of 160 raster images representing 16 geographical areas, out of which 323 time-patches (96 natural and 227 artificial) and 562 patches (335 natural and 227 artificial) were extracted. Assuming a training on 50 epochs, DCNNs were effectively trained on more than 16 000 augmented time-patches, or more than 28 000 augmented patches.

7) Implementation Details: Because standard DCNNs have an output of dimension 1000, we need to add a final fully connected layer from dimension 1000 to dimension 2 (either natural or artificial oil slick). The output is a vector of two scalars with values proportional to the confidence that the model has the set of patches that belongs to the class. Those scalars can be normalized to give probabilities. In our case, the prediction of the network is the class corresponding to the biggest scalar.

We added a dropout layer [55] with probability 0.3 before the fully connected layer. We also used L2 regularization (weight-decay = 0.1) to force the model to learn simple patterns rather than complex ones. We used AdaBelief optimizer with a learning rate lr = 10-4. This set of hyperparameters was the set with best accuracy on the validation set when computing Grid Search on the following combinations: Dropout layer in $[0, 0.1, 0.2, 0.3, 0.4, 0.5]$, L2 regularization in $[0, 10^{-6}, 10^{-4}, 10^{-2}, 0.1, 0.2, 0.5, 1]$, and learning rate in $[0, 10^{-6}, 10^{-4}, 10^{-2}, 0.1, 0.2, 0.5, 1]$.

The dataset being slightly imbalanced, we used binary cross-entropy loss with weighted coefficients corresponding to the proportion of every label in the training dataset. Formally, if the binary cross entropy loss takes y as the true label (either 0 or 1), and the prediction of the network is p (a continuous prediction from 0 to 1), then $\text{BCEL}(y, p) = -y\log(p) + (1-y)\log(1-p)$. In our case, since there are 2.36 times more patches labeled as containing an artificial oil slick than patches containing a natural oil seepage, we used $\text{Weighted_BCEL}(y, p) = -y\log(p) + 2.36(1-y)\log(1-p)$ where $y = 0$ corresponds to natural seepage, and $y = 1$ to artificial oil slicks.

III. Results

A. Comparison of the Baseline Network Architectures

The quantitative results obtained on the training and test dataset of the baseline with several different architectures are shown in Table II. Note that the corresponding results using obtained using Random Forest, SVM, and K-nearest neighbor after hyperparameters optimization were 81.2%, 69.4%, and 63.7% on the train set, and 66.4%, 67.3%, and 63.3% on the test set, respectively. All baseline DCCNs we tested therefore obtained clearly better accuracies and generalization properties than these models. As suggested in Section II-B2, we believe that this is due to the ability of DCNNs to efficiently manage semi-local information in the images. Other information about the various DCNNs used is available in the Appendix. In these results, Resnet152 [32] outperforms the other algorithms in terms of training loss and accuracy. This is expected behavior as this model is the one containing the largest amount of layers. It can therefore fit more than other models to the training observations. However, when observing the test loss, AUC, and accuracy, we understand that ResNet152 overfitted the training data. It therefore appears as having poorer generalization abilities than ResNet18 in our study. Resnet18 was the best-performing algorithm on the test set (unseen data) both in terms of accuracy, AUC, and loss, which means it was able to capture the complexity of the task without overfitting. This justifies the choice of using Resnet18 as the baseline architecture for the rest of this article.

TABLE II
Results Obtained Using Various Deep Convolutional Neural Networks for the Baseline Task

DCNN	Best train accuracy	Best val accuracy
VGG11	80.2%	79.9%
VGG19BN	92.7%	81.1%
RESNET18	93.7%	**82.3%**
RESNET152	96.0%	79.6%

B. Comparison of Individual Features as Inputs of the Resnet18 Architecture

In Table III, we now compare individually the individual statistical features of Section II-B3 obtained on the baseline Resnet18 architecture with one input channel. It turns out the three best extractors were *variance*, *norm + power average*, and *average* both in terms of training loss and of validation accuracy. They will then be used as the three meta-input channels of the *statistical extraction* time-method in Section III-C.

C. Comparison of Different Temporal Integration Strategies on the Resnet18 Architecture

We finally compare in Table IV all time-methods that exploit time-patches. The accuracy confusion matrix obtained on the test set using the *shortcutting constrained convolution* time-method (which performed best) is also given in Table V.

IV. Discussion

We will focus on discussing results for the task of oil seepage/oil spills distinction to stress out our main contribution, and general results for oil seepage detection and oil spills detection are also presented in the appendix section.

As depicted in the introduction section, the state-of-the-art for SAR image distinction often consisted of deep convolutional neural networks that performed both the feature extraction and the classification. Our baseline that does not

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TABLE III
INDIVIDUAL RESULTS OBTAINED USING DIFFERENT OF STATISTICAL EXTRACTORS

Statistical extractor	Best train accuracy	Best val accuracy
Power average (d)	81.4%	58.2%
Min (e)	82.0%	62.2%
Variance (f)	93.1%	**73.0%**
Norm+power average (g)	90.1%	**64.4%**
Average (h)	92.7%	**63.0%**

TABLE IV
DETECTION PERFORMANCE OF THE TESTED TIME-METHODS WITH RESNET18. THE TIME-METHODS THAT OUTPERFORM THE BASELINE ARE IN BOLD FONT. 150 EPOCHS WERE REQUIRED FOR THE CONVERGENCE OF SHORTCUTTING METHODS, WHILE 50 EPOCHS WAS ENOUGH WITH OTHER METHODS

	Best train accuracy	Best test accuracy
Constained convolution	87.7%	73.5%
Full convolution	83.3%	77.7%
Statistical feature extraction	99.3%	**89.2%**
Sorting	99.0%	**85.5%**
Averaging	89.7%	80.5%
Shortcutting constrained convolution	95.7%	**92.3%**
Shortcutting full convolution	95.7%	**88.6%**
Shortcutting sorting full convolution	92.0%	**83.8%**
Shortcutting sorting constrained convolution	71.7%	63.0%

TABLE V
CONFUSION MATRICES OBTAINED ON BALANCED TEST DATA WITH AND WITHOUT THE *Shortcutting Constrained Convolution Time-Method* (S.C.C.)

Baseline ResNet18	Pred. Natural seepage	Pred. Artificial spill
True Natural seepage	80.3%	19.7%
True Artificial spill	14.7%	85.3%

ResNet18 with S.C.C.	Pred. Natural seepage	Pred. Artificial spill
True Natural seepage	90.0%	10.0%
True Artificial spill	4.4%	95.6%

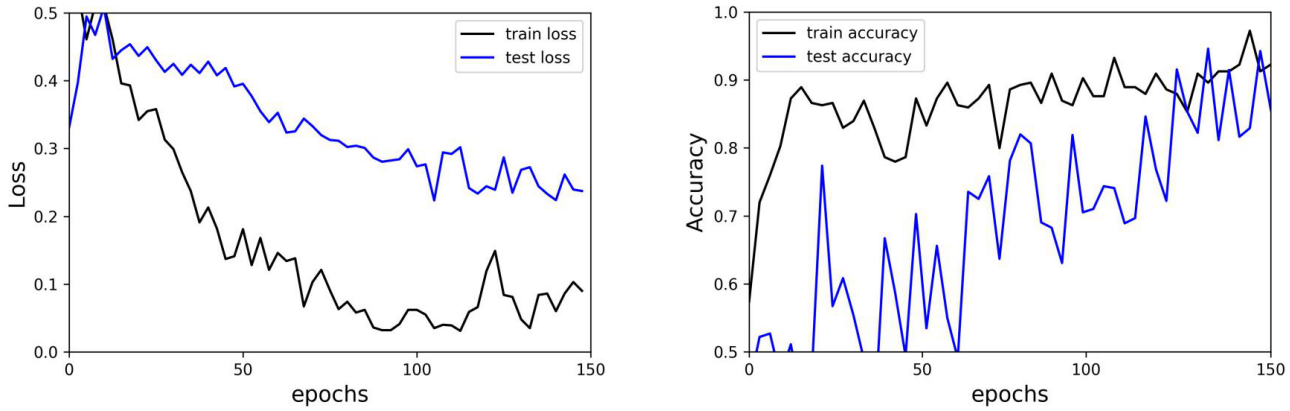

Fig. 7. Evolution of the (Left) loss and the (Right) accuracy when training the shortcutting constrained convolutional time model.

use temporal data follows a similar approach and reaches 82% which is very close to the state of the art in natural and anthropogenic oil slick distinction: [49] reaches 81% overall accuracy using a simple machine learning classifier (random forest) and a selected subset of 400 manually extracted features from SAR images. However, we observe that our baseline reaches a lower accuracy than the state-of-the-art for other tasks such as oil detection where [38] reaches 94% overall accuracy using a similar approach to our baseline, or the state-of-the-art for oil spill segmentation: 91% intersection over

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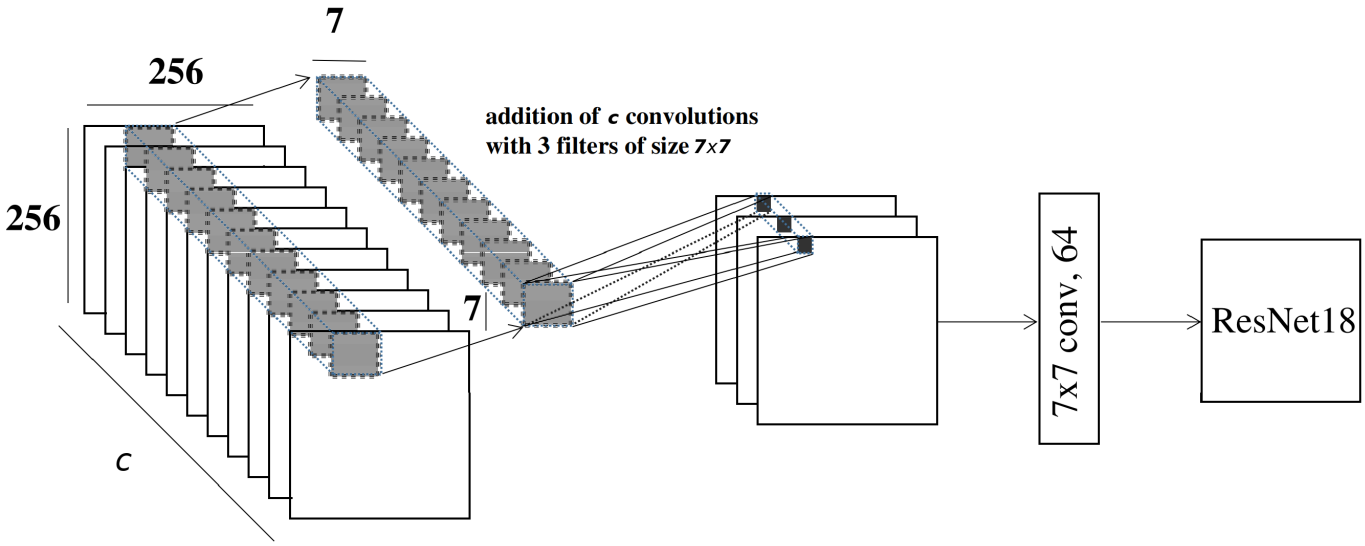

Fig. 8. Detailed temporal integration model for the *constrained convolution time-method*.

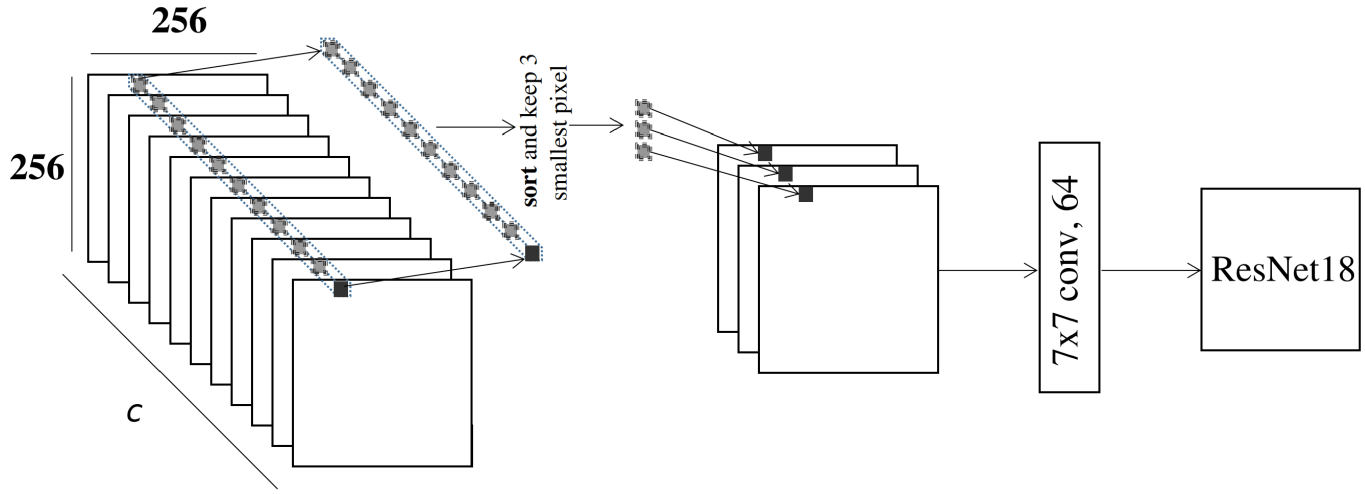

Fig. 9. Detailed temporal integration model for the *sorting time-method*.

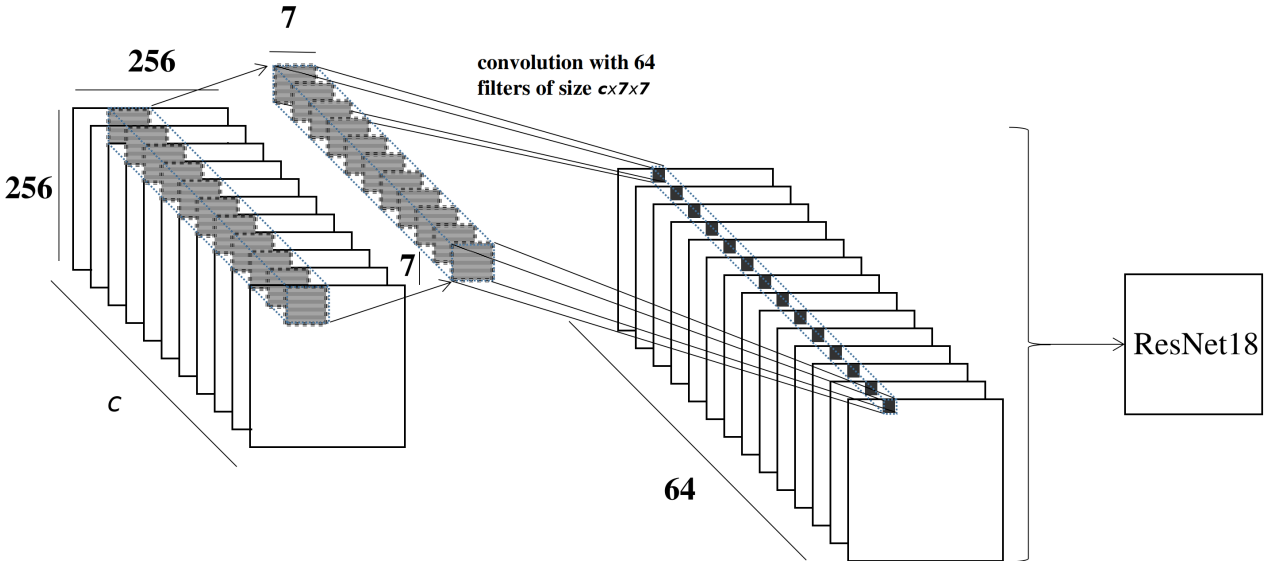

Fig. 10. Detailed temporal integration model for the *shortcutting full convolution time-method*.

union accuracy [47]. It seems that distinguishing natural seepages and artificial oil spills is a harder task. However, when considering temporal information, our contribution reaches 92% overall accuracy which pushes the accuracy for this task

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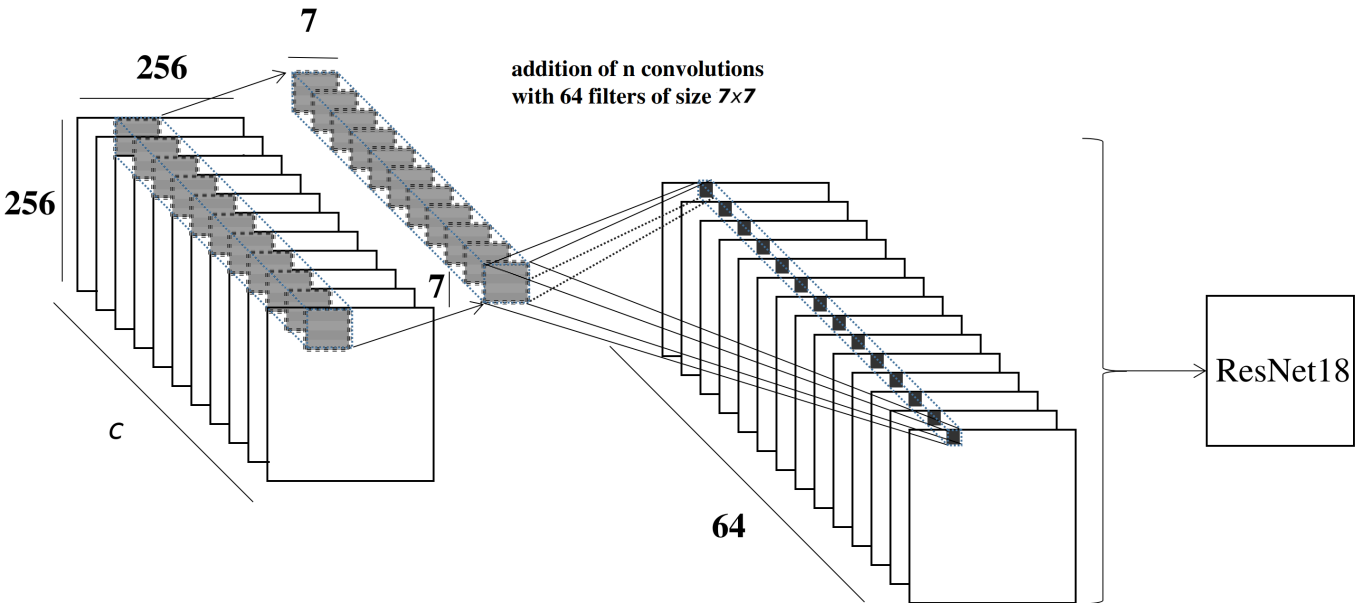

Fig. 11. Detailed temporal integration model for the *shortcutting constrained convolution time-method*.

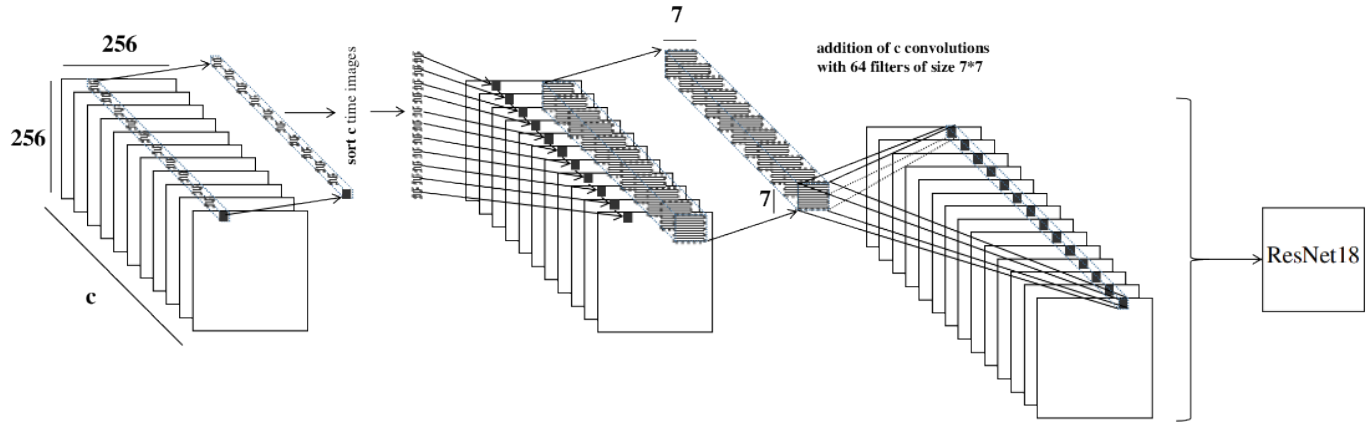

Fig. 12. Detailed temporal integration model for the *sorting + shortcutting full convolution time-method*.

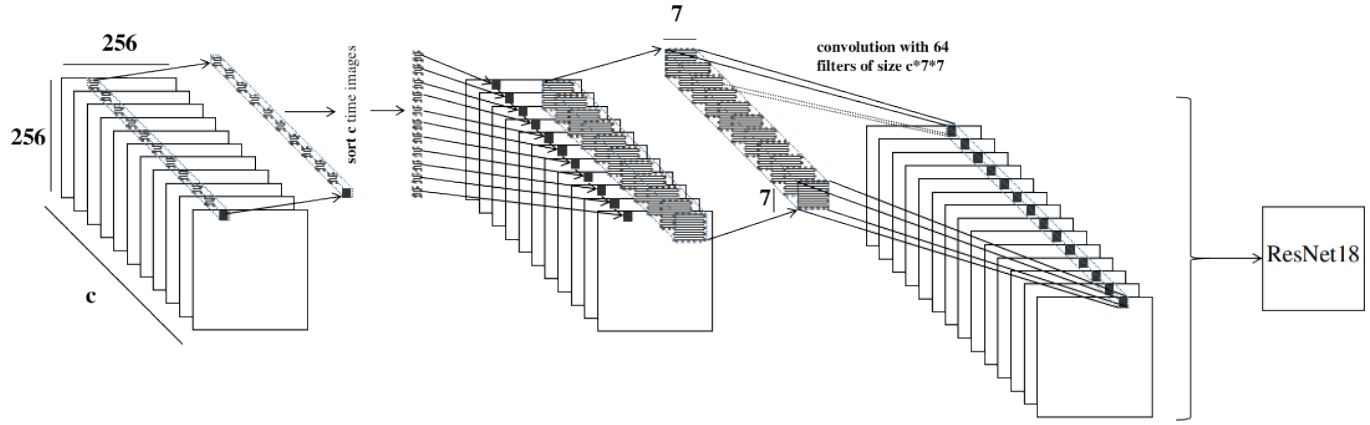

Fig. 13. Detailed temporal integration model for the *sorting + shortcutting constrained convolution time-method*.

to an equivalent level to the state-of-the-art accuracy for oil spill detection: 94%, or oil spill segmentation: 91%.

Other trends can be noticed when comparing the time-methods: Nonlinear time-methods such as *sorting* or *statistical extractor* time-methods first outperform linear time-methods like *convolution filters*, probably because they allow a more meaningful information extraction. Shortcutting the first layer of the network also provides better results, probably because the information is not squeezed into a bottleneck with only three dimensions [56].

It can also be remarked that *constrained convolutions* outperform *full convolutions* for *normal* and *shortcutting* time-methods, but not for the *sorting + shortcutting* time-method. This is expected because full convolutions apply

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TABLE VI
CONFUSION MATRIX ON TEST DATASET WITH ADDITIONAL CLASS "CLEAR SEA OR LOOK ALIKE" USING THE "SHORTCUTTING CONSTRAINED CONVOLUTION" TIME-METHOD

	Pred. Natural seepage	Pred. Artificial spill	Pred. Clear sea or look-alikes
True Natural seepage	82.6%	13.0%	4.4%
True Artificial spill	7.4%	90.5%	2.1%
True Clear sea or look-alikes	1.3%	4.4%	94.3%

TABLE VII
MISCELLANEOUS ABOUT THE VARIOUS DEEP CONVOLUTIONAL NEURAL NETWORKS USED FOR THE BASELINE TASK

DCNN	**Number of parameters**	**number of layers**	**Best test AUC**
VGG11	133M	11	74.5 %
VGG19BN	144M	19	76.3 %
RESNET18	11M	18	**77.9** %
RESNET152	60M	152	75.1 %

a different filter to every channel, which does not match our task where the order of every channel is not relevant. This overfitting from *full convolution* time-methods has been reduced by our time-permutation augmentation method described, yet *full convolution* methods tend to outperform *constrained convolutions* during the training (overfitting) and perform worse on test data. On the contrary, the *shortcutting* + *sorting* + *constrained convolution* time-method performs worse than the *shortcutting* + *sorting* + *full* time-method, which is expected as well because in this case, the order of the channels is relevant as they have been bitwise ordered, and it is relevant to apply a different convolutional filter on every channel.

Extracting time-patches drastically reduced the size of the dataset. The maximum number of samples in the dataset must be divided by at least c. However, the size of our dataset was reduced by a bigger factor $3 \times c$ because we privileged a simpler image preprocessing that only clustered together images with a shared covering area. The optimal preprocessing strategy is finer grained as it consists of clustering patches from all images with a shared area, instead of only exploiting images from the same image cluster. The DCNN used, ResNet18, being the simplest in terms of the number of parameters probably helped to regularize the overfitting effects. We can however observe important accuracy fluctuations in Fig. 7.

This effect is lighter when considering losses values because the loss considers the confidence that the model has about the patch belonging to both classes rather than only the final prediction. Nevertheless, using the temporal information for natural and anthropic oil slick distinction proved to be a very insightful feature as it permitted us to outperform the previous state-of-the-art despite a small dataset.

Overfitting might also originate from the huge dimensionality of our time-patches: Since time-patches have c channels instead of 3, there are $(256 * 256)^{c-3}$ times more possibilities of inputs. The compression to three channels for nonshortcutting time-method reduces the number of possibilities back to $(256 * 256)^3$ as usual. Time-methods shortcutting the first layer are hence expected to experience more overfitting because these approaches might struggle with the huge dimensional input space.

Note finally that our method can also be used on multiclass classification. For instance, we extended our work from *oil spill/oil seepage discrimination* to *oil spill/oil seepage/clear sea or lookalikes* detection, and a micro-averaged F1 score of 89.1% was obtained. More detailed results are given in the Appendix B, and more specifically in the confusion matrix Table VI.

Further work could include a finer extraction of the time-patches by clustering at the patch level instead of the raster image level. Other time-methods or extraction strategies could also be included and compared. Finally, our work could easily be generalized to a segmentation task, this would only require replacing the classifying DCNN with a segmenting DCNN such as Deeplabv3+, and to take the union of segmented patches of a time-patch to create the label of the time-patch.

V. CONCLUSION

In this article, we used SAR images to detect and inventory oil slicks presence in the ocean domain. Our contribution was to explore the use of novel block layers to take advantage of the temporal redundancy of natural seepages in specific locations of the ocean, when discriminating natural seepages from artificial slicks. The experimental protocol we used to explore the efficiency of the tested neural-network architectures was close to operational conditions, where the amount of annotated data used to train the neural-networks, is often limited because of the large amount of time they demand from experts. The first experimental steps (baselines) which correspond to classical discrimination methods allow us to get similar prediction accuracies compared to the literature using standard (mono-temporal) neural-networks. More interestingly, the integration of the temporal information enabled us to reach 92.3% overall accuracy over 1600 augmented test time-patches (see Table IV), while the best accuracy obtained using standard images was 82.3% (see Table II). This means that the average error rate was divided by a factor of more than two using the strategy proposed in this article. The comparison of different techniques to integrate the temporal dimension showed that using nonlinear methods to compress the information or removing the information bottleneck provided higher accuracy scores. The results presented in this article suggest that considering the temporal occurrences of

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seeps is a meaningful feature to be considered for oil spill discrimination of satellite SAR imagery.

APPENDIX

A. Illustration of Different Temporal Integration Block Layers

We illustrate in Figs. 8–13 the detailed architecture of the different temporal integration block layers defined and tested in this manuscript.

B. Additional Test With Look Alikes

Although our methodology aims to discriminate oil spills from oil seepages, it may be interesting to add a third class *clear sea or look alikes*. This would make our method more generic for detection and not only for discrimination. We therefore test our best method when adding the additional class *clear sea or look alikes*. We add samples of this class such that the ratio of train and test data is as follows: natural oil seepage/artificial oil spill/clear sea or look alikes = 0.42/1/50. We use the same hyper parameters as for the case with two classes except that we assign the weights of the weighted cross entropy loss according to our new class proportions. We report a test macro F1 score of 89.1%, and a test micro F1 score of 94.1%. Results are given in Table VI. We observe that the accuracy for the two original classes decreases, but the added class is relatively easy to classify, which increases the F1 scores.

C. Additional Information About the Deep Convolutional Neural Networks

We present in Table VII miscellaneous information about the various deep convolutional neural networks used for the baseline task.

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